

Subject: ML slides pack - association rules by Claudio Sartori

Course: Artificial Intelligence - LM

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TIPS

- w.r.t means "with respect to"

Association rules → discovering co-occurrences in a market basket

Given a set of commercial transactions, find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction

Example of association rules

- $\{\text{Diaper}\} \rightarrow \{\text{Beer}\}$
- $\{\text{Bread, Milk}\} \rightarrow \{\text{Coke, Eggs}\}$
- $\{\text{Beer, Bread}\} \rightarrow \{\text{Milk}\}$

Implication means co-occurrence, not causality!

The implication of Association Rules is different from that of logic (boolean): **it can be true with some level of truth**

Market basket transactions

TID	Items
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk

Definition: frequent Itemset

Itemset

A collection of one or more items

Example: $\{\text{Bread, Diaper, Milk}\}$

k-itemset

An itemset that contains k items

Support count (σ)

Frequency of occurrence of an itemset

E.g. $\sigma(\{\text{Bread, Diaper, Milk}\}) = 2$

Support

Fraction of transactions that contain an itemset

E.g. support of $\{\text{Bread, Diaper, Milk}\} = \frac{2}{5}$

Frequent Itemset

An itemset whose support is greater than or equal to a minsup threshold

TID	Items
1	Bread, Milk
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4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk

Definition: association rule

Association Rule

An expression of the form $A \rightarrow C$, where A and C are itemsets

A = Antecedent and C = Consequent

Example: $\{\text{Diaper, Milk}\} \rightarrow \{\text{Beer}\}$

Rule Evaluation Metrics

Support (sup)

Fraction of the N transactions that contain both A and C

Confidence (conf)

Measures how often all the items in C appear in transactions that contain A

$$\text{support}(A \rightarrow C) = \frac{\sigma(A \cup C)}{N}$$

$$\text{confidence}(A \rightarrow C) = \frac{\sigma(A \cup C)}{\sigma(A)}$$

Example:

- For rule $\{\text{Diaper, Milk}\} \rightarrow \{\text{Beer}\}$:
 - $\sigma(\{\text{Beer, Diaper, Milk}\}) = 2$
 - $\sigma(\{\text{Diaper, Milk}\}) = 2$
 - $N = 5$
 - $\text{support} = \frac{2}{5} = 0.4$
 - $\text{confidence} = \frac{2}{2} = 1.0$

TID	Items
1	Bread, Milk
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Why support and confidence?

- Rules with low support can be generated by random associations
- Rules with low confidence are not really reliable
- Nevertheless a rule with relatively low support but high confidence can represent an uncommon but interesting phenomenon

Association rule mining task

Given a set of transactions $\mathcal{N}$, the goal of association rule mining is to find all rules having:

- **support** $\geq$ minsup threshold
- **confidence** $\geq$ minconf threshold

Brute-force approach:

1. List all possible association rules
2. Compute the support and confidence for each rule
3. Prune rules that fail the minsup and minconf thresholds

This is computationally prohibitive!

Mining association rules

Example of rules from itemset **{Beer, Diaper, Milk}**:

Rule	Support (s)	Confidence (c)
{Diaper, Milk} $\rightarrow$ {Beer}	0.4	0.67
{Beer, Milk} $\rightarrow$ {Diaper}	0.4	1.0
{Beer, Diaper} $\rightarrow$ {Milk}	0.4	0.67
{Beer} $\rightarrow$ {Diaper, Milk}	0.4	0.67
{Diaper} $\rightarrow$ {Beer, Milk}	0.4	0.5
{Milk} $\rightarrow$ {Beer, Diaper}	0.4	0.5

Key observation:

All the rules above are binary partitions of the same itemset: **{Beer, Diaper, Milk}**

Important property:

Rules originating from the same itemset have **identical support** but can have **different confidence**

Implication:

We may decouple the support and confidence requirements

Two-step approach for efficient mining

1. Frequent itemset generation

Generate all itemsets whose support is greater than minsup

2. Rule generation

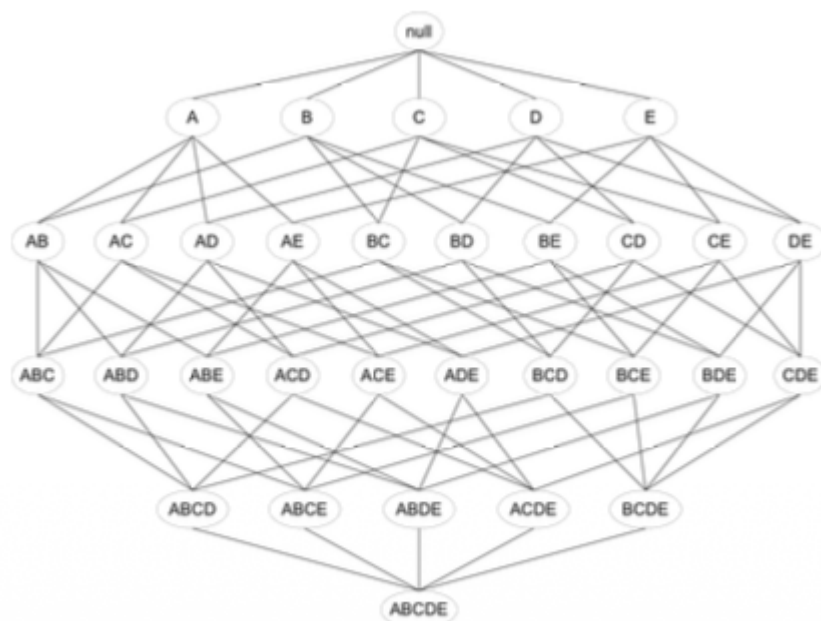
Generate high confidence rules from each frequent itemset, where each rule is a binary partitioning of a frequent itemset

Note:

Even with this decoupling, **frequent itemset generation is still computationally expensive**

Frequent itemset generation

Given D items, there are $M = 2^D$ possible candidate itemsets



Brute-force approach

- Each itemset in the lattice is a candidate frequent itemset
- Count the support of each candidate by scanning the database
- Match each transaction against every candidate

Complexity: $O(N \cdot W \cdot M) \implies$ Expensive!

TID	Items
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk

(Where N = number of transactions, W = average transaction width, M = number of candidate itemsets)

Brute-force computational complexity (OPTIONAL!!!)

Given D unique items:

$$\text{Total number of itemsets} = 2^D$$

Total number of possible association rules:

$$R = \sum_{k=1}^{D-1} \left[\binom{D}{k} \times \sum_{j=1}^{D-k} \binom{D-k}{j} \right] = 3^D - 2^{D+1} + 1$$

1. Count the number of ways to create an itemset that forms the left-hand side of the rule
2. For each size k itemset selected for the left-hand side, count the number of ways to choose the remaining $D - k$ items to form the right-hand side of the rule

Going deeper:

- Choose k of the D items for the left-hand side of the rule: there are $\binom{D}{k}$ ways to do this
- There are $\binom{D-k}{j}$ ways to choose the right-hand side of the rule, where $1 \leq j \leq D - k$
- The double summation derives from the two points above
- The binomial theorem states that $\sum_{i=0}^n \binom{n}{i} x^i = (1 + x)^n$
- Using the theorem for $x = 1$ and $x = 2$ leads to the final result (pay attention to the starting value of the summation)

Frequent itemset generation strategies

Goal: Reduce computational complexity

1. Reduce the number of candidates M

- Complete search: $M = 2^D$
- Use pruning techniques to reduce M

2. Reduce the number of comparisons $N \cdot M$

- Use efficient data structures to store the candidates or transactions
- No need to match every candidate against every transaction

The Apriori principle

Apriori principle:

If an itemset is frequent, then all of its subsets must also be frequent

Why does this hold?

Due to the anti-monotone property of the support measure:

$$\forall X, Y : (X \subseteq Y) \implies \text{support}(X) \geq \text{support}(Y)$$

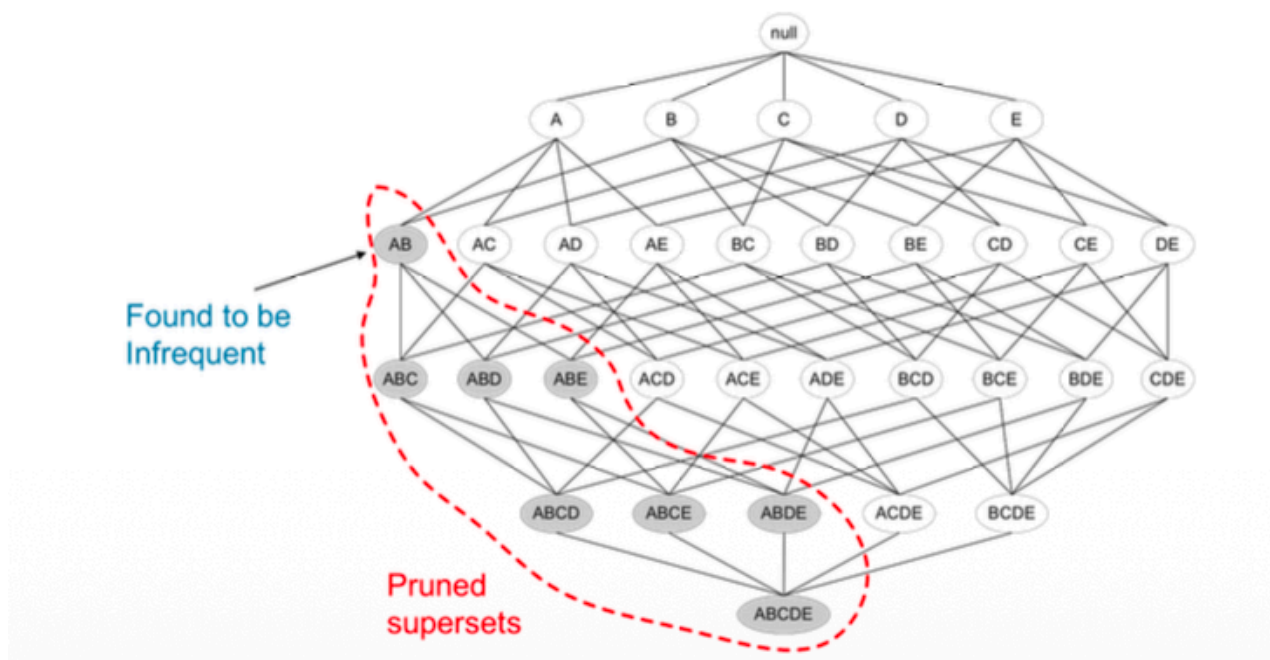
Key property:

The support of an itemset never exceeds the support of its subsets

This is known as the **anti-monotone property** of support

Consequences for pruning:

- If any subset of an itemset is **infrequent**, the itemset itself must also be **infrequent**
- We can prune candidate itemsets early by checking their subsets
- This dramatically reduces the search space in frequent itemset mining



- **Apriori algorithm - candidate generation**

Definitions

- C_k : candidate itemsets of size k
- L_k : frequent itemsets of size k
- $\text{subset}_k(c)$: set of the subsets of c with k elements

Candidate generation – join step

Let L_k be represented as a table with k columns where each row is a frequent itemset.

Let the items in each row of L_k be in **lexicographic order**.

C_{k+1} is generated by a **self join** of L_k :

```

INSERT INTO C_{k+1}
SELECT p.item1, p.item2, ..., p.item_k, q.item_k
FROM L_k AS p, L_k AS q
WHERE p.item1 = q.item1
      AND p.item2 = q.item2
      ...
      AND p.item_{k-1} = q.item_{k-1}
      AND p.item_k < q.item_k;

```

How it works:

- Take two frequent k -itemsets from L_k
- They must share the first $k - 1$ items (ensures proper merging)
- The last items must be different, with $p.item_k < q.item_k$ (avoid duplicates)
- This creates a candidate $(k + 1)$ -itemset by combining them

Origin of the name "Apriori"

Level-wise computation:

- The level is the cardinality of the itemsets under evaluation
- The evaluations at level k use the **prior knowledge** acquired for the previous levels to reduce the search space

Factors affecting complexity

1. Choice of minimum support threshold

- Lowering support threshold $\rightarrow$ more frequent itemsets
- This reduces pruning effectiveness and increases maximum length of frequent itemsets
- **Number of database scans = maximum length of frequent itemsets + 1**

2. Dimensionality (number of items) of the dataset

- More items $\rightarrow$ more space needed to store support counts
- If number of frequent items increases $\rightarrow$ both computation and I/O costs increase

3. Size of database

- Since Apriori makes multiple passes, runtime increases with number of transactions

4. Average transaction width

- Denser datasets $\rightarrow$ wider transactions
- Wider transactions $\rightarrow$ increase maximum length of frequent itemsets
- **Why?** Number of subsets in a transaction grows exponentially with its width

Confidence

The confidence of a rule can be computed from the supports:

$$\text{conf}(A \rightarrow C) = \frac{\text{support}(A \cup C)}{\text{support}(A)}$$

Key insight:

For confidence-based pruning of rules, it is sufficient to know the supports of frequent itemsets

Rule generation

Step 1: Generate candidate rules

Given a frequent itemset L , find all non-empty subsets $f \subset L$ such that:

$$\text{conf}(f \rightarrow (L - f)) \geq \text{minconf}$$

Example: From $\{\text{Beer}, \text{Diaper}, \text{Milk}\}$, the possible rules are:

- Beer, Diaper $\rightarrow$ Milk
- Beer $\rightarrow$ Diaper, Milk
- Beer, Milk $\rightarrow$ Diaper
- Milk $\rightarrow$ Beer, Diaper
- Diaper, Milk $\rightarrow$ Beer
- Diaper $\rightarrow$ Beer, Milk

If $|L| = k$, there are $2^k - 2$ candidate rules
(Rules $L \rightarrow \emptyset$ and $\emptyset \rightarrow L$ can be ignored)

Efficient rule generation

Challenge

In general, confidence does **not** have an anti-monotone property:

$$\text{conf}(ABC \rightarrow D) \text{ can be larger or smaller than } \text{conf}(AB \rightarrow D)$$

Key observation for same itemset

Consider rules generated from the same itemset, e.g., $I = \{A, B, C, D\} \in L$:

$$\text{conf}(ABC \rightarrow D) \geq \text{conf}(AB \rightarrow CD) \geq \text{conf}(A \rightarrow BCD)$$

Important property:

Confidence of rules generated from the same itemset is **anti-monotone** with respect to the number of items on the RHS of the rule

It **decreases** when we move an item from the left-hand side to the right-hand side

Rule generation in Apriori

Candidate rule generation

A candidate rule is generated by **merging two rules** that share the same prefix in the rule consequent.

Example:

- Rule 1: $CD \rightarrow AB$
- Rule 2: $BD \rightarrow AC$

Join these to produce candidate rule: $D \rightarrow ABC$

Pruning rule candidates

Prune rule $D \rightarrow ABC$ if its subset $AD \rightarrow BC$ does not have high confidence.

Why?

Based on the anti-monotone property of confidence within the same itemset:
If a more specific rule (with more items on LHS) has low confidence,
a more general rule (with fewer items on LHS) cannot have higher confidence.

Pattern evaluation

The problem with association rules

Association rule algorithms tend to produce **too many rules**:

- Many are uninteresting or redundant

Example of redundancy:

- $\{A, B, C\} \rightarrow \{D\}$ and $\{A, B\} \rightarrow \{D\}$ have same support and confidence
- The first rule doesn't add meaningful information beyond the second

Interestingness measures

- Can be used to **prune/rank** derived patterns
- Help identify truly interesting associations

Note:

In the original formulation of association rules, **support** and **confidence** are the only measures used

Computing interestingness measures

Given a rule $A \rightarrow C$, the information needed to compute rule interestingness can be obtained from a **contingency table**:

	C	$\neg C$	Total
A	f_{11}	f_{10}	$f_{1\cdot}$
$\neg A$	f_{01}	f_{00}	$f_{0\cdot}$
Total	$f_{\cdot 1}$	$f_{\cdot 0}$	

The elements of the contingency table are the basis for most interestingness measures

Drawback of confidence

Example contingency table:

	Coffee	¬ Coffee	Total
Tea	15	5	20
¬ Tea	75	5	80
Total	90	10	100

$$\text{conf}(\text{Tea} \rightarrow \text{Coffee}) = \frac{\text{supp}(\text{Tea}, \text{Coffee})}{\text{supp}(\text{Tea})} = \frac{15}{20} = 0.75$$

- $P(\text{Coffee}) = 0.9$
- $P(\text{Coffee} \mid \neg \text{Tea}) = \frac{75}{80} = 0.9375$

Problem: Despite high confidence of `Tea → Coffee`, the absence of Tea actually **increases** the probability of Coffee!
For this rule, confidence is misleading.

Statistical independence

Example: Population of 1000 students

- 600 know how to swim (S)
- 700 know how to bike (B)
- 420 know both ($S \wedge B$)

$$P(S \wedge B) = \frac{420}{1000} = 0.42$$

$$P(S) \times P(B) = 0.6 \times 0.7 = 0.42$$

Statistical independence: $P(S \wedge B) = P(S) \times P(B)$

- $P(S \wedge B) > P(S) \times P(B) \rightarrow$ **Positively correlated**
- $P(S \wedge B) < P(S) \times P(B) \rightarrow$ **Negatively correlated**

Lift

$$\text{lift}(A \rightarrow C) = \frac{\text{conf}(A \rightarrow C)}{\text{supp}(C)} = \frac{P(A, C)}{P(A) \times P(C)}$$

- Evaluates to **1** for independence
- **Insensitive to rule direction**
- Ratio of true cases w.r.t. independence

Leverage

$$\text{leverage}(A \rightarrow C) = P(A, C) - P(A) \times P(C) = \text{supp}(A \cup C) - \text{supp}(A) \times \text{supp}(C)$$

- Evaluates to **0** for independence

- **Insensitive to rule direction**
- Number of additional cases w.r.t. independence

Conviction

$$\text{conv}(A \rightarrow C) = \frac{1 - \text{supp}(C)}{1 - \text{conf}(A \rightarrow C)} = \frac{P(A)(1 - P(C))}{P(A) - P(A, C)}$$

- Infinite if the rule is always true
- **Sensitive to rule direction**
- Ratio of expected frequency that A occurs without C (if independent) divided by observed frequency of incorrect predictions
- Also called **novelty**

Intuition about measures

- **Higher support** → rule applies to more records
- **Higher confidence** → chance that the rule is true for some record is higher
- **Higher lift** → chance that the rule is just a coincidence is lower
- **Higher conviction** → the rule is violated less often than it would be if antecedent and consequent were independent

Comparison of measures

	C1	$\overline{C1}$	
A1	88	5	93
$\overline{A1}$	5	2	7
	93	7	100

Rule ($A1 \Rightarrow C1$)

$$\text{conf} = 0.88/0.93 = 0.946$$

$$\text{lift} = 0.88/(0.93 * 0.93) = 1.017$$

$$\text{leve} = 0.88 - 0.93 * 0.93 = 0.015$$

$$\text{conv} = (1 - 0.93)/(1 - 0.946) = 1.302$$

A high confidence rule can have small lift if both sides are very frequent

	C2	$\overline{C2}$	
A2	2	5	7
$\overline{A2}$	5	88	93
	7	93	100

Rule ($A2 \Rightarrow C2$)

$$\text{conf} = 0.02/0.07 = 0.286$$

$$\text{lift} = 0.02/(0.07 * 0.07) = 4.082$$

$$\text{leve} = 0.02 - 0.07 * 0.07 = 0.015$$

$$\text{conv} = (1 - 0.07)/(1 - 0.286) = 1.302$$

A low confidence rule can have high lift if both sides are very infrequent

Lift: Relative association strength

Practical meaning

How many times the presence of the consequent is **inflated** if the antecedent is present.

Measures departure from statistical independence

Key limitations

- **Inflated for rare items** (can give misleadingly high values for infrequent items)
- **No notion of absolute business impact** (only relative measure)

Leverage: Absolute incremental effect

$$\text{Leverage}(A \rightarrow B) = P(A \cap B) - P(A)P(B)$$

Practical meaning

If leverage = 0.04: "4% of all transactions contain the pair because of the association"

Business interpretation

In 1M transactions, this rule explains **40,000 extra joint purchases** beyond what would be expected by chance.

Measures **incremental volume**, not correlation

Conviction: Rule reliability

$$\text{Conviction}(A \rightarrow B) = \frac{P(A)P(\neg B)}{P(A \cap \neg B)}$$

Practical meaning

How strongly the rule **avoids being wrong** (penalizes cases where A occurs but B does not)

Visual intuition: Confidence vs Conviction

	Confidence	Conviction
What it measures	$P(B A)$	$\frac{\text{expected failures under independence}}{\text{observed failures}}$
Perspective	Looks only at successes	Compares failures to independence baseline
Intuition	"When I recommend, how often am I right?"	"How surprising is it when the recommendation fails?"
Warning sign	-	High confidence + low conviction $\rightarrow$ frequent false positives

When lift is misleading

Counterexample 1: Rare items

Assume:

- $P(A) = 0.01, P(B) = 0.01, P(A \cap B) = 0.005$

$$\text{Lift} = \frac{0.005}{0.0001} = 50$$

Problem: Extremely high lift but only 0.5% of transactions affected

Business reality: Rule is statistically strong but commercially irrelevant

Counterexample 2: Popular items

Assume:

- $P(A) = 0.80$, $P(B) = 0.70$, $P(A \cap B) = 0.60$

$$\text{Lift} \approx 1.07$$

Observation: Lift suggests weak association

Insight: Yet 60% of users satisfy the rule!

Lift **undervalues** rules involving popular items

Mapping metrics to recommender KPIs

Metric	Measures	Related KPI
Lift	Relative predictiveness	CTR uplift
Leverage	Incremental volume	Revenue lift
Conviction	Error avoidance	Precision@K
Confidence	Success frequency	Hit Rate
Support	Coverage	Catalog utilization

Practical guidance

Use lift for discovery and filtering

- Identifies statistically interesting associations
- Good for finding novel patterns

Use leverage for revenue prioritization

- Quantifies absolute business impact
- Directly translates to potential revenue

Use conviction for recommendation reliability

- Measures how consistently the rule holds
- Important for user trust and experience

Rule of thumb:

High lift $\wedge$ high leverage $\wedge$ high conviction $\rightarrow$ production-ready rule

Conclusion on measures

- Many measures exist beyond the four presented
- **Confidence** is usually the base tool
- Other measures can test and filter confidence results
- **No single perfect measure** – use combination based on business goals

Multidimensional association rules

Comparison: Mono- vs multi-dimensional

	Mono-dimensional	Multi-dimensional
Event	Transaction	Tuple
Event description	Items A, B, C are together in a transaction	Attribute A has value a, B has value b, C has value c in a tuple

Example: Sensor data

Look for rules like: `CO = high AND Tin Oxide = high → Titanium = high`

Use case: If one sensor is unavailable, guess its qualitative value from others.

Handling quantitative attributes

Too many distinct values for direct multi/mono transformation → need **discretization**.

Approaches:

- Equifrequency binning
- Mono-dimensional clustering for optimal covering

Key insight:

Association rules can involve items at different qualitative levels after discretization

Multilevel association rules

The challenge

A real market basket database can include **tens of thousands** of distinct items.

Need to find trade-off between:

- **General reasoning** (higher-level patterns)
- **Detailed reasoning** (specific item combinations)

Key: Choose the right level of abstraction

Solution: Concept hierarchy

Common background knowledge is the organization of items into a **hierarchy of concepts**:

- Can be easily coded in transactions
- Helps choose the right level of abstraction

Support in multilevel association rules

From specialized to general

Example: `(apple → milk) → (fruit → dairy)`

Support increases (in general)

New rules can become interesting at higher levels

From general to specialized

Example: $(\text{fruit} \rightarrow \text{dairy}) \rightarrow (\text{apple} \rightarrow \text{milk})$

Support decreases (in general)

Support can go under the threshold at lower levels

Confidence in multilevel association rules

A level change can influence confidence **in any direction** (increase, decrease, or remain similar).

Redundancy rule:

If a specialized rule has (approximately) the same confidence as the general one, then it is **redundant**

Example

- 1000 transactions
- 80 with milk and bread
- 114 with milk
- 20 with low-fat milk and bread
- 28 with low-fat milk

Rule a: $\text{milk} \rightarrow \text{bread}$

- Support = 8%
- Confidence = 70%

Rule b: $\text{low-fat milk} \rightarrow \text{bread}$

- Support = 2%
- Confidence = 71%

Since rule b) has almost the same confidence as rule a) and is a descendant, **rule b) is redundant**

Mining multilevel association rules

Approach: Top-down exploration

1. Look for frequent itemsets at **each level of abstraction**, top-down
2. Each level requires a new run of the rule discovery algorithm
3. **Decrease the support threshold** in lower levels (to account for decreasing support)

This hierarchical approach helps manage the complexity while maintaining meaningful patterns at appropriate abstraction levels.